Supplementary Material
LEGALEDIT: A Dissociation Diagnostic for Legal Meaning Preservation Metrics

Gaurab Baral
University of Cincinnati
baralgb@mail.uc.edu

This document collects the tables, figures and secondary analyses referred to in LEGALEDIT: A Dissociation Diagnostic for Legal Meaning Preservation Metrics. Numbers with an S prefix are defined here; section, table and figure numbers without one refer to the main paper. Everything here comes from the same pipeline and the same released result files.

A Supplementary Tables and Figures

The judge as a sixth annotator, and the combination attempts. Because the judge emits on the annotators' scale it can be placed inside the panel rather than only beside the metrics. Its mean rating is 9.11, against a human panel spanning 4.34 to 7.61, so it is more lenient than any human and far less dispersed. Yet its mean correlation with the five annotators is 0.544, higher than the mean human–human correlation of 0.462. Adding it as a sixth coder nonetheless *lowers* ordinal α from 0.325 to 0.300: it tracks the consensus without belonging to the distribution. Against its own five samples it is far more self-consistent than the humans are with each other ($\alpha = 0.689$ versus 0.325), and averaging the samples is not cosmetic, since correlation against the human mean climbs from 0.611 at $K = 1$ to 0.719 at $K = 5$. Its rubric-level explanations show the pattern Section 6.3 of the main paper describes: on identical pairs it reports an error in 0.0% of samples, and on LEGALEDIT it reports some error in 78.1% and names *incohérence*, the correct type for a meaning-reversing edit, in 54.6%. On the capability and combination controls: substituting a reasoning model (`deepseek-reasoner`) under the identical rubric and items gives $r = 0.580$ against 0.619 and margin 0.320 against 0.425, so the shortfall is not a matter of the model thinking harder; and of six rules for merging the judge with bidirectional NLI, every calibration fitted on the training split alone and the diagnostic never entering the fit, none beats both parents on both axes. The best, a weighted blend at $w = 0.74$, reaches $r = 0.762$ and margin 0.574, gains smaller than the redraw-to-redraw variation. We report this as a negative result rather than proposing a metric. The full protocol is 7,080 calls costing $0.99.

Margin by perturbation rule. Table S6 gives the per-rule breakdown that Section 5 of the main paper summarises, averaged over metrics. Two details behind the party-swap figure quoted there are worth stating. The two directions of the swap draw mean margins of 0.046 and -0.005 on 11 and 3 items respectively, which is why we pool them rather than report either cell on its own: over all 14 party-swap items the seven similarity metrics spend 0.020 of their range, with a per-metric spread of 0.006–0.036, against 0.029 for those same seven metrics across all rules. The contrast is milder on this restricted basis than the all-metric figures in Section 5 of the main paper suggest, because the similarity family has almost no range to spend on any rule. We also tested whether the residual ordering across rules is explained by the size of the edit in characters, and found no relationship ($r = 0.25$ over the 12 rules, 95% CI $[-0.66, 0.73]$). With 12 rules of unequal frequency the ordering is not itself well determined, so we report it descriptively; the robust observation is the ceiling on detectability, not the ranking.

B Training Variants in Full

The protocol so far audits scorers that already exist. It also has implications for how one would fit a new one, and this section tests two of them. The first follows from Section 6.1 of the main paper: the gold label is the mean of five ratings whose median spread is 7 points, so fitting a scalar regressor to that mean asks the model to predict a quantity none of the annotators asserted, and rewards predicting the

Metric	r	ρ	RMSE_{raw}	RMSE_{cal}	over%
Unsupervised, French-capable					
BERTScore-CamemBERTv2	$0.291_{\pm 0.09}$	0.326	2.62	2.36	49.3
BERTScore-CamemBERTv2-Recall	$0.352_{\pm 0.12}$	0.398	2.59	2.30	47.2
BERTScore-FlauBERT	$0.482_{\pm 0.08}$	0.459	2.16	2.12	49.4
LaBSE	$0.409_{\pm 0.06}$	0.402	2.33	2.21	47.2
Sentence-CamemBERT	$0.296_{\pm 0.05}$	0.287	2.68	2.31	49.3
Para-mUSE-mpnet	$0.252_{\pm 0.08}$	0.266	2.88	2.35	47.2
mE5-base	$0.358_{\pm 0.08}$	0.356	2.52	2.26	48.0
NLI-fwd (no hallucination)	$0.368_{\pm 0.06}$	0.352	3.85	2.25	49.3
NLI-bwd (no omission)	$0.340_{\pm 0.09}$	0.355	3.70	2.28	46.9
NLI-bidirectional (min)	$0.395_{\pm 0.07}$	0.405	3.74	2.22	46.6
Trivial surface features					
$-\lvert\,\lvert s\rvert - \lvert o\rvert\,\rvert$ (length difference)	$0.641_{\pm 0.04}$	0.624	3.25	1.85	47.0
Token Jaccard	$0.303_{\pm 0.06}$	0.331	3.34	2.34	48.0
Length ratio	$0.580_{\pm 0.05}$	0.560	3.43	1.97	49.1
Surface-Ridge (10 features, supervised)	$0.621_{\pm 0.05}$	0.588	1.89	1.89	45.8
Prompted LLM judge					
LLM-judge (deepseek-chat, k=5)	$0.755_{\pm 0.05}$	0.753	2.93	1.57	43.9
Human ceiling (one annotator)	**0.597**	0.620	–	3.27	–

Table S1: Calibrated comparison on the FRJUDGE test split, mean over 10 seeds. RMSE_{raw} is the decimal-scaling protocol in current use; RMSE_{cal} fits an isotonic map on the training split. r and ρ are both computed after calibration; being monotone, the map leaves ρ almost unchanged (at most 0.014) but shifts r by up to 0.074.

centre. The second follows from Section 3 of the main paper: if a diagnostic identifies a deficiency, the obvious response is to train against it.

The seven configurations share an encoder (CamemBERTv2-base), splits, schedule and seeds, and differ only in the target and the loss.

- **Scalar** (the conventional objective): MSE against the mean rating.
- **Distributional**: a ten-way head trained with KL divergence against the empirical rating histogram, following Uma et al. [5] and Peterson et al. [4]. The reported score is the distribution's expectation, so it is directly comparable, but the model additionally exposes its spread.
- **Annotator-aware**: five regression heads, one per annotator, averaged at inference. This lets the model represent the lenient and strict populations of Section 6.1 of the main paper separately instead of averaging them away.
- **Quantile**: a pinball loss [3] at $\tau = 0.25$. The asymmetry is deliberate. A metric that *overshoots* the human rating is permissive of bad simplifications, while one that undershoots is merely strict, and in a legal setting those errors are not symmetric; a quantile objective encodes that preference in the loss rather than hoping it emerges.
- **Multi-task**: the scalar head plus an auxiliary characterization head, trained against the majority vote of the five annotators, which spans 17 of the rubric's 18 classes (the eighteenth never carries a majority). The characterization labels are collected in FRJUDGE and then discarded; since the published rubric has annotators characterize a clause *before* scoring it, the label plausibly carries information about the score.
- **DA**: the scalar objective with the identical/unrelated augmentation in current use added to the training mixture, unchanged.
- **DA+LegalEdit**: **DA** plus minimally-edited negatives generated from FRJUDGE originals by the rules of Section 4 of the main paper, disjoint from the diagnostic's own sentences.

Evaluation protocol. Correlation and RMSE are computed *only* on the human-annotated test split, never on a split containing augmentation. This matters: adding identical pairs (label 10) and unrelated pairs (label 1) to the evaluation set inflates label variance at the extremes, which raises Pearson r for every metric without any improvement in judgment. A table that does not separate the two shows the signature clearly, in that a metric's r and its RMSE improve and worsen together rather than moving as one. Augmentation here affects training only; the

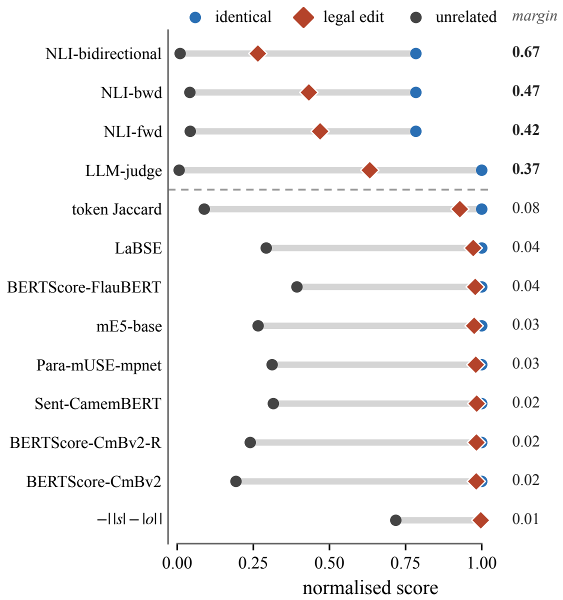

Figure S1: LEGALEDIT. For each metric the grey bar spans its score on an unrelated pair (dark circle) to its score on an identical pair (blue circle): its full working range. The red diamond is where it places a sentence whose legal force has been reversed by a single word. A metric that measured legal meaning would put the diamond at the left end. Every similarity-based metric puts it at the right, hard against the identical pair. Only the four scorers above the dashed rule, the three NLI variants and the prompted judge, move it appreciably. Metric names are abbreviated; Table 1 of the main paper gives them in full, with the margin column repeated at the right here.

peut → doit	*tier 1*
o: La Société peut refuser une indemnité, en réduire le montant, en suspendre ou en cesser le paiement dans les cas suivants: 1.	
s': La Société **doit** refuser une indemnité, en réduire le montant, en suspendre ou en cesser le paiement dans les cas suivants: 1.	
doit → peut	*tier 1*
o: Le candidat doit toutefois réussir les examens de compétence visés à l'article 67 pour obtenir un permis de conduire pour motocyclette.	
s': Le candidat **peut** toutefois réussir les examens de compétence visés à l'article 67 pour obtenir un permis de conduire pour motocyclette.	
drop *ne . . . pas*	*tier 1*
o: Les feux visés au paragraphe 2° ne sont pas requis pour tout véhicule dont la largeur excède 2,03 mètres.	
s': Les feux visés au paragraphe 2° sont requis pour tout véhicule dont la largeur excède 2,03 mètres.	
avant ↔ après	*tier 1*
o: Elle n'est pas due avant le septième jour qui suit celui de l'accident, sauf dans le cas prévu au troisième alinéa de l'article 57.	
s': Elle n'est pas due **après** le septième jour qui suit celui de l'accident, sauf dans le cas prévu au troisième alinéa de l'article 57.	
tous les → certains	*tier 1*
o: Un agent de la paix peut enlever ou faire enlever aux frais du propriétaire toute chose utilisée en contravention au présent article.	
s': Un agent de la paix peut enlever ou faire enlever aux frais du propriétaire **certaine** chose utilisée en contravention au présent article.	

Table S2: LEGALEDIT samples, one per perturbation rule, with the edited span in bold. Each remains fluent and statutory in register, shares almost all of its tokens with the original, and states a different rule of law.

three diagnostic regimes are reported separately as pass rates.

Disagreement-aware objectives buy stability, not accuracy. Table S7 reports the grid, and the first result is a negative one: no training target moves the correlation by more than the scalar baseline's own spread across seeds (± 0.082), which is itself smaller than the 0.73 gap Section 6.2 of the main paper showed to be resolvable on an 89-item split. The differences that survive are in the other columns. The distributional head is markedly steadier than the scalar head it replaces (± 0.029 against ± 0.082 over the same five seeds), reaches the best RMSE in the table (1.99), and roughly doubles the rank correlation ($\rho = 0.578$ against 0.280), which is what one expects from a target carrying the shape of the rating distribution and not only its mean. The quantile head does precisely what it was asked to do: it overshoots the human label on 9.9% of test items against 25.4% for the scalar head, the lowest rate of any model here. The risk asymmetry a legal setting calls for can therefore be written into the loss rather than hoped for, and it costs nothing in correlation.

Trained constraints do not cross a change of document. Every model in Table S7 scores 0% on the identical-pair check, the two augmented ones included, although identical pairs are in their training mixture; the augmented models pass the unrelated check on 77% and 69% of probes and the un-augmented models on none. The asymmetry is the one our probe design predicts. Our unrelated probes are drawn from the same two statutes the training mixture draws its unrelated pairs from, so passing that check is an in-distribution result; our identical probes are statutory sentences while the mixture's identical pairs are FRJUDGE clauses, so passing that one would require generalising across source documents, and no model does. The same

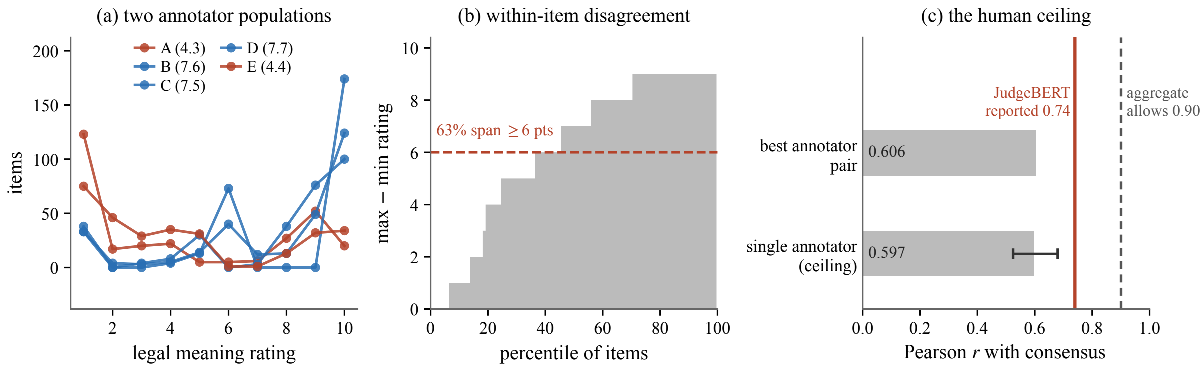

Figure S2: Disagreement in FRJUDGE. (a) The five annotators form two populations: B, C, D (blue) rate legal preservation about three points higher than A and E (red) across the whole corpus. (b) Sorted within-item range; the median item spans 7 points of a ten-point scale. (c) A single annotator predicts the consensus of the other four at $r = 0.597$ (bar; whiskers give the range over the five held-out annotators), and the best-agreeing pair reaches $r = 0.606$.

Dimension	Krippendorff's α		Reported
	nominal	appropriate	prev.
Simplicity level	0.183	0.183	0.18
Characterization	0.552	0.552	0.55
Legal meaning	0.104	**0.325**	0.10

Table S3: Krippendorff's α on FRJUDGE. Legal meaning is an ordinal ten-point scale, for which the nominal coefficient is not the appropriate measure. *prev.* is the value reported by Beauchemin et al. [1].

Held-out	r (95% CI)	ρ	RMSE
A	0.524 [0.44, 0.60]	0.551	3.93
B	0.569 [0.48, 0.65]	0.586	3.05
C	0.680 [0.60, 0.75]	0.725	2.75
D	0.601 [0.52, 0.67]	0.557	3.08
E	0.612 [0.55, 0.67]	0.679	3.56
Mean (ceiling)	**0.597**	0.620	3.27
Best pair (C-D)	0.606	–	–
Mean pairwise	0.462	–	–

Table S4: Leave-one-annotator-out ceiling on FRJUDGE ($n = 297$). A single expert predicts the consensus of the other four at $r = 0.597$.

holds for the diagnostic: DA+LEGALEDIT, trained on minimally-edited negatives, reaches a margin of 0.051 against 0.67 for untrained bidirectional NLI, its negatives generated from FRJUDGE originals and its probes from statutes. Training on a constraint buys satisfaction of it in distribution and nothing at all across a change of source.

The same separation explains the augmented mixtures' correlations, which are the clearest effect in the table: DA loses 0.350 against the unaugmented baseline and DA+LEGALEDIT 0.420, on five seeds out of five in both cases, and DA varies by ± 0.254 across seeds, reaching $r = -0.188$ on one of them.

One column of Table S7 should be read with care. LEGALEDIT false-acceptance is the share of flipped pairs still scored ≥ 7, so it is confounded with where a model puts its predictions overall. The distributional head's 69% reflects a head that scores high in general (its identical probes average 7.22), and the 0% of the scalar heads reflects models that score low on everything, not models that reject the edit.

C Williams Tests Against the Length Baseline

The test split holds 89 items, at which size the smallest correlation reliably distinguishable from $r = 0.46$ at 80% power is 0.73. The ten unsupervised metrics and the length feature of Table S1 correlate with the *same* human label on the *same* 297 items, and with each other, so the comparison is dependent and a two-sample test on Fisher-transformed correlations would be anticonservative. Table S8 reports the Williams test [2], which corrects for that dependence.

D What Survives a Change of Document

A metric meta-evaluation that never changes source document cannot distinguish legal sensitivity from familiarity with a form. We therefore add an out-of-domain split. FRJUDGE clauses come from identifiable source documents, which we recover from the raw export. Random splitting lets clauses from the same insurance form appear in both training and test, and insurance forms

Configuration	r	edit	margin	discrim.%	asym.
`chat`, rubric (FR) *[primary]*	0.687	6.25	0.421	82.0	3.24
`chat`, rubric (EN)	0.727	4.84	0.573	88.0	3.63
`chat`, bare instruction	0.758	3.83	0.687	89.0	4.03
`chat`, rubric, K=1	0.619	6.22	0.425	79.0	3.34
`reasoner`, rubric, K=1	0.580	7.12	0.320	87.0	3.79

Table S5: The prompted judge across models and prompt wordings, on a fixed 100-edit subsample of LEGALEDIT so the reasoner stays affordable. The first row is the FRJUDGE rubric in French, the second the same rubric in English over the same French text, and the third a bare instruction with the error taxonomy removed. The three wordings are drawn at K=2 and the two capability rows at K=1; within a row every column comes from the same draw, so the wordings are comparable with each other and the two models with each other, but a row is not comparable with the K=5 figures quoted elsewhere. *asym.* is the mean score on a *peut→doit* edit minus that on a *doit→peut* edit; both reverse the law.

Perturbation	n	mean margin	worst
drop *ne ... pas*	21	0.333	0.032
avant ↔ après	9	0.272	0.006
amount × 10	21	0.256	0.025
doit → peut	82	0.241	0.015
sauf si → si	10	0.225	0.014
peut → doit	111	0.167	0.018
tous les → certains	42	0.153	0.017
et → ou	27	0.069	0.007
ou → et	26	0.069	0.004
assureur → assuré	11	0.046	0.007
automobile ↔ véhicule	10	0.027	-0.025
assuré → assureur	3	-0.005	-0.086
All	373	**0.155**	–

Table S6: LEGALEDIT margin by perturbation rule, averaged over the metrics in Table 1 of the main paper. *worst metric* is the lowest margin any single metric assigns that rule. A margin of 1 would mean the metric treats a legally flipped sentence exactly as it treats an unrelated one.

are highly templated. We compare the standard random split against a document-grouped split in which no source document straddles the boundary (Table S9); the difference estimates how much of the reported performance is form-specific. We rerun three of the seven configurations of Table S7 on that split: the scalar baseline and the two augmented mixtures. The remaining four differ from the baseline in what they are trained to predict rather than in what they can exploit about a source form, so the leakage they are exposed to is the same.

The corpus makes that harder than it sounds. Its 297 pairs come from only 21 source forms and the three largest hold 71% of them, so the obvious construction, sweeping the forms in random order and cutting at the cumulative 60/70% marks, lets a single large form overshoot a boundary and starve the split behind it. On seeds 42–46 that procedure returns test splits of 16, 87, 30, 0 and 88 pairs, one of them empty and one far too small to carry a correlation. We therefore choose the form-to-split assignment by a short local search over total deviation from the target sizes. Thousands of assignments hit 178/30/89 exactly, so each seed still selects a genuinely different partition while every split matches the random-split sizes of Table S7 pair for pair, and the two tables differ only in which forms are held out. One limitation is structural rather than incidental: the largest form is 37% of the corpus, larger than the 30% test quota, so it always falls in training and the grouped test set is always drawn from the remaining 188 pairs.

The comparison does not resolve. The scalar baseline's correlation falls from 0.542 on the random split to 0.454 on the grouped one, a drop of 0.088 that is in the direction leakage would predict. It does not survive inspection. The per-seed differences are $+0.169$, -0.097, -0.416, -0.182 and $+0.086$: two of the five seeds *improve* under the harder split, and the grouped spread (± 0.143) is wider than the gap itself. The two augmented mixtures move by -0.002 and $+0.026$, which is to say not at all. As a control, the untrained NLI reference is fitted to nothing and should be indifferent to how the corpus is partitioned; it moves from 0.395 to 0.391, confirming that the two splits are otherwise comparable and that the variance above is the models' and not the split's.

We therefore do not claim that document-level leakage inflates the correlations of Section B, and we hold our own evidence to the standard Section 6.2 of the main paper applies elsewhere. What we can say is narrower and, we think, more useful to anyone planning a successor corpus: FRJUDGE is too small and too concentrated to answer the

Model	r	ρ	RMSE	over%	ident.	unrel.	LegalEdit	
					pass	pass	false-acc.	margin
JudgeBERT-Scalar (baseline)	$0.542_{\pm 0.08}$	0.280	2.48	25.4	0	0	0	–
JudgeBERT-Dist (soft labels)	$0.572_{\pm 0.03}$	0.578	1.99	42.9	0	0	69	0.015
JudgeBERT-Annot (5 heads)	$0.534_{\pm 0.05}$	0.585	3.47	12.6	0	0	0	–
JudgeBERT-Quantile (tau=.25)	$0.553_{\pm 0.06}$	0.377	3.28	9.9	0	0	0	–
JudgeBERT-MT (+charact.)	$0.519_{\pm 0.06}$	0.232	2.51	26.3	0	0	0	–
JudgeBERT-DA	$0.192_{\pm 0.25}$	0.112	2.31	42.5	0	77	18	0.019
JudgeBERT-DA+LegalEdit	$0.122_{\pm 0.10}$	0.174	2.60	35.3	0	69	0	0.051
NLI-bidirectional (untrained)	0.395	0.405	2.22	46.6	1	–	–	*0.670*
Human ceiling	**0.597**	0.620	3.27	–	–	–	–	–

Table S7: Trained metrics, mean over 5 seeds, evaluated on the human-annotated test split only. *ident.* and *unrel.* are the two sanity checks in current use; LegalEdit false-acceptance is the share of legally flipped pairs still scored ≥ 7 ("accurate" under the published rubric). The margin is the quantity defined in Section 3 of the main paper. It is shown as '–' where a model places the identical and unrelated probes within one point of each other, because a model with no working range has none to spend on the edit.

Neural metric m	$r(m, h)$	$r(m, \ell)$	t	p
BERTScore-CamemBERTv2	0.344	0.349	5.05	<0.001
BERTScore-CamemBERTv2-Recall	0.396	0.443	4.43	<0.001
BERTScore-FlauBERT	0.484	0.389	2.61	0.010
LaBSE	0.424	0.362	3.66	<0.001
Sentence-CamemBERT	0.222	0.148	6.33	<0.001
Para-mUSE-mpnet	0.225	0.159	6.32	<0.001
mE5-base	0.318	0.316	5.40	<0.001
NLI-fwd (no hallucination)	0.320	0.388	5.66	<0.001
NLI-bwd (no omission)	0.307	0.258	5.36	<0.001
NLI-bidirectional (min)	0.365	0.350	4.68	<0.001

Table S8: Williams tests against the trivial length baseline $\ell = -\lvert\, |s| - |o| \,\rvert$, which reaches $r = 0.613$ with the human label h over all 297 items. $r(m, \ell)$ is each metric's correlation with the length feature itself, the dependence the test corrects for. Positive t favours ℓ.

question. Its grouped test split is always drawn from the 188 pairs the largest form leaves behind, at which size the seed-to-seed variation swamps the effect being measured. A corpus intended to support an out-of-domain claim needs its documents spread far more evenly than three forms holding 71% of the data, and this is the aspect of the design we would change first.

E Stratifying by Length Change

We further break the test predictions down by the magnitude of the length change between original and simplification. If a metric's correlation is concentrated in the buckets where the simplification is much shorter, it is detecting deletion rather than legal change, the hypothesis raised in Section 6.2 of the main paper. Table S10 reports Pearson r inside each tercile of $\lvert\, |s| - |o| \,\rvert$. Within a tercile the length signal is largely held constant, so a metric that retains its correlation there is using something other than deletion; one whose correlation collapses was reading length all along. This is a weaker form of the same dissociation logic as Section 3 of the main paper: it holds a confound approximately fixed by conditioning rather than exactly fixed by construction.

The result qualifies Section 6.2 of the main paper in a way that is worth stating carefully, because it cuts against our own argument. The length feature's correlation is overwhelmingly a *between*-tercile effect: pooled over the corpus it reaches 0.613, but inside the small and medium terciles it falls to 0.243 and 0.165. Once items of comparable length change are compared with each other, knowing the length change tells you little. It recovers to 0.585 only in the tercile where the simplification is much shorter than the original, which is exactly where deletion is available to be detected. The neural baselines have the opposite profile: in the medium tercile, where the length feature retains almost nothing, BERTScore over FlauBERT holds 0.449 against a pooled 0.484 and LaBSE rises to

Model	r	ρ	RMSE	over%	ident. pass	unrel. pass	LegalEdit false-acc.	LegalEdit margin
JudgeBERT-Scalar (baseline)	$0.454_{\pm 0.14}$	0.282	2.49	24.0	0	0	0	–
JudgeBERT-DA	$0.190_{\pm 0.21}$	0.140	2.63	33.0	0	100	9	0.004
JudgeBERT-DA+LegalEdit	$0.149_{\pm 0.13}$	0.319	2.65	28.1	0	49	0	0.030
NLI-bidirectional (untrained)	0.391	0.401	2.20	45.7	1	–	–	*0.670*
Human ceiling	**0.597**	0.620	3.27	–	–	–	–	–

Table S9: Document-grouped split: no source form appears in both training and test. Columns are as in Table S7, and the splits are size-matched to it (178/30/89). The untrained NLI reference is recomputed on the grouped splits rather than carried over, since a corpus-level correlation is not comparable to a document-grouped one; the human ceiling is a property of the annotations and does not depend on the split.

Metric	r within tercile small	medium	large	all
$-\big\vert\,\vert s\vert - \vert o\vert\,\big\vert$	0.243	0.165	0.585	0.613
BERTScore-FlauBERT	0.198	0.449	0.413	0.484
LaBSE	0.174	0.532	0.240	0.424
NLI-bidirectional	0.058	0.269	0.291	0.365
Surface-Ridge	0.118	0.305	0.540	0.620
JudgeBERT-Scalar	0.042	0.149	0.602	0.496

Table S10: Correlation with the human label *within* terciles of the absolute length change, pooling test predictions over seeds. Restricting to a tercile removes most of the length variance; a metric that measures legal meaning should degrade far less than one that measures deletion.

0.532 against a pooled 0.424. The supervised surface ridge tracks the raw length feature bucket for bucket, which is what one would expect if it is largely a length model with extra terms. The small tercile is hard for everything, which we read as a floor effect rather than as a result: when the two sentences are close to the same length, none of these signals has much to go on.

The honest reading is therefore narrower than "legal meaning preservation is length preservation". The corpus-level ranking is dominated by length, and any comparison that omits a length control will attribute to a metric what it should attribute to that confound. But the neural metrics are not *only* reading length. Where the length signal is weakest, they are the ones that hold up, keeping or exceeding the correlation they had over the whole corpus. Whatever they are reading in that stratum, it is not the length change. What Section 5 of the main paper adds is that the residual signal they retain is still not legal sensitivity, since the same metrics move almost not at all when the law is changed and the length is held exactly fixed. The two analyses answer different questions, and a metric has to survive both.

F Reproducibility

Every number we report is produced by a single pipeline that runs end to end on a laptop; the largest model involved is CamemBERTv2-base (112M parameters) and no GPU cluster is required. We release the experiment code at `https://github.com/Nyvora-Vision-Labs/LEGALEDIT`: the reconstruction of FrJudge from the raw export (which recovers annotator identity and source document, both dropped from the public JSONL), the document-grouped splits, the LegalEdit generator and scoring harness, and the result files behind every table.

One caveat on exact reproduction: our training runs use Apple MPS kernels, which are not bitwise deterministic, so a fixed seed reproduces a run only approximately: repeat runs of the scalar configuration at seed 42 differ in test-set r by several hundredths, which is comparable to the spread across seeds. We therefore report means and standard deviations over five seeds and treat any difference smaller than that spread as unresolved, the same discipline Section 6.2 of the main paper asks of everyone else, applied to ourselves. Anyone reproducing this on CUDA should expect tighter agreement and correspondingly narrower error bars.

Two further details matter for anyone reproducing the diagnostic. The first is what the probes share with training, which differs by regime and which we state precisely because conflating the two is one of the failure modes the protocol is meant to prevent. The *identical* probe is built from statutory sentences (the Quebec Automobile Insurance Act and the Highway Safety Code) while the identical pairs in the -DA training mixture are built from FrJudge's insurance-form clauses, so a model passes that probe only by generalising across source documents. The LEGALEDIT probe is likewise generated from statutory sentences, while the minimally-edited negatives added by the DA+LEGALEDIT variant are generated from FRJUDGE originals: same perturbation rules, disjoint sentences. The *unrelated* probe is the weakest of the three in this respect: both it and the unrelated pairs in the training mixture are drawn from the same two statutes, under different random seeds, so a high unrelated pass rate for a -DA model should be read as in-distribution and we do not treat it as evidence of generalisation. Second, the ordinal Krippendorff coefficient we report is computed with the same library and the same reliability matrix as the nominal one, so the difference between 0.104 and 0.325 is attributable to the distance function alone.

References

[1]. David Beauchemin, Michelle Albert-Rochette, Richard Khoury, and Pierre-Luc Déziel. 2025. JUDGEBERT: Assessing legal meaning preservation between sentences. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 92–118.

[2]. Yvette Graham and Timothy Baldwin. 2014. Testing for significance of increased correlation with human judgment. In *EMNLP*.

[3]. Roger Koenker and Gilbert Bassett Jr. 1978. Regression quantiles. *Econometrica*, 46(1):33–50.

[4]. Joshua C Peterson, Ruairidh M Battleday, Thomas L Griffiths, and Olga Russakovsky. 2019. Human uncertainty makes classification more robust. In *ICCV*.

[5]. Alexandra N Uma, Tommaso Fornaciari, Dirk Hovy, Silviu Paun, Barbara Plank, and Massimo Poesio. 2021. Learning from disagreement: A survey. *JAIR*, 72:1385–1470.